

Safe Controller Synthesis for UAV Trajectories under Stochastic Disturbances

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Abstract—This work addresses the challenge of ensuring safe Unmanned Aerial Vehicle (UAV) operation in the presence of stochastic disturbances, which are inherent in real-world environments. Existing approaches to UAV safety primarily rely on offline risk assessment or simulation-based validation and therefore lack formal guarantees during real-time operation under uncertainty. To overcome this limitation, a probabilistic safety verification framework is proposed that integrates stochastic reachability analysis with deterministic trajectory planning.

The method models UAV dynamics as a stochastic system subject to sub-Gaussian disturbances and derives high-confidence bounds on trajectory deviations using a set-erosion formulation. These bounds are incorporated into the planning process by inflating obstacles in the configuration space, effectively transforming the original stochastic problem into a deterministic one with modified constraints. This enables the use of standard graph-based planning algorithms while implicitly enforcing probabilistic safety guarantees.

The framework is implemented in a high-fidelity Gazebo-ArduPilot simulation environment with realistic wind disturbances modeled using Dryden and von Kármán turbulence models. Experimental results demonstrate that the proposed approach achieves bounded trajectory tracking error, exhibits contractive closed-loop behavior, and maintains safe operation under significant environmental disturbances. The UAV successfully follows planned trajectories and completes missions while remaining within probabilistically safe regions.

Overall, the proposed method provides a computationally efficient and practically applicable solution for integrating formal probabilistic safety guarantees into UAV trajectory planning. It contributes toward bridging the gap between theoretical safety verification methods and real-world autonomous UAV operation in uncertain environments.

I. INTRODUCTION

UAVs are increasingly deployed in a wide range of safety-critical and mission-critical applications, including infrastructure inspection, logistics, environmental monitoring, disaster response, precision agriculture, and emerging urban air mobility systems [1]. As outlined in recent comprehensive studies on UAV path planning and autonomous operation, the rapid expansion of application domains has significantly increased the complexity of operational environments, which are often dynamic, partially unknown, and subject to strict safety, energy, and computational constraints [2].

In these real-world settings, ensuring safe and reliable autonomous operation under uncertainty is a fundamental challenge. UAVs are inherently affected by stochastic disturbances such as wind gusts, atmospheric turbulence, sensor noise, localization errors, and modeling inaccuracies [2]. These

effects become particularly critical in complex scenarios such as extreme weather conditions, GPS-denied environments, high-speed flight, and multi-agent systems, where assumptions of perfect sensing or static environments no longer hold [1]. In such cases, even small deviations in state estimation or control can propagate into significant trajectory errors, increasing the risk of collision or mission failure.

Existing safety assurance approaches for UAVs are primarily based on offline risk assessment frameworks, simulation-based validation, and rule-based operational constraints [2]. Regulatory methods such as the Specific Operations Risk Assessment (SORA) framework provide structured procedures for hazard classification and mitigation design [3, 4]. However, these approaches are inherently conservative and do not provide formal guarantees on system behavior during real-time operation under uncertainty. Similarly, simulation-based testing allows for evaluation under a wide range of scenarios, but it is fundamentally limited to pre-defined conditions and cannot guarantee robustness against unseen disturbances [2].

In parallel, significant progress has been made in trajectory planning methods for UAVs. Classical approaches include graph-based search algorithms such as Dijkstra and A*, sampling-based methods such as RRT and PRM, and optimization-based techniques including evolutionary algorithms, particle swarm optimization, and ant colony optimization. More recently, machine learning and deep reinforcement learning methods have been introduced to improve adaptivity and real-time decision-making in dynamic environments. These approaches have demonstrated strong performance in complex scenarios such as urban navigation, multi-UAV coordination, and obstacle-rich environments [1].

Despite these advances, most existing trajectory planning methods treat safety as either a heuristic objective or enforce it under deterministic assumptions. As a result, they often lack robustness against stochastic disturbances, particularly in highly uncertain environments where probabilistic effects dominate system behavior.

To address these challenges, probabilistic reachability analysis has emerged as a promising theoretical foundation for reasoning about system behavior under uncertainty. Such methods allow the propagation of stochastic disturbances through non-linear dynamics and enable the computation of probabilistic bounds on trajectory deviations [5, 6, 7]. However, despite their strong theoretical guarantees, these approaches are rarely integrated into practical UAV planning systems due to their

computational complexity and limited compatibility with real-time trajectory generation frameworks.

This paper addresses this gap by proposing a probabilistic safety verification framework for UAV trajectory planning under stochastic disturbances. The core idea is to integrate probabilistic reachability analysis with deterministic trajectory planning through a set-erosion formulation. In this framework, high-confidence bounds on trajectory deviations are derived from stochastic system dynamics and environmental uncertainty. These bounds are then used to inflate obstacles in the configuration space, effectively transforming the original stochastic planning problem into a deterministic one with modified geometry. This allows conventional planning algorithms to operate efficiently while implicitly guaranteeing probabilistic safety.

The framework is evaluated in a Gazebo–ArduPilot simulation environment under realistic stochastic wind conditions modeled using Dryden and von Kármán turbulence models. The results demonstrate that the proposed method achieves bounded trajectory tracking error, maintains closed-loop stability under uncertainty, and ensures successful mission execution in challenging environmental conditions.

The remainder of this paper is structured as follows. Section II reviews related work in UAV safety, trajectory planning, and uncertainty-aware control. Section III presents the theoretical foundation of probabilistic safety verification and disturbance modeling. Section IV describes the simulation environment and implementation details. Section V presents experimental results and performance evaluation. Finally, Sections VI and VII conclude the paper and discuss directions for future research.

II. RELATED WORK

Ensuring the safe operation of UAVs is traditionally addressed through systematic safety assessment procedures conducted during the design and certification phases. Existing safety assessment approaches can be categorized into model-based, analytical, data-driven, experimental, and hybrid methods. Among these, model-based and analytical approaches dominate current practice [2]. Analytical frameworks such as SORA and MEDUSA provide structured methodologies for hazard identification, risk evaluation, and mitigation, and are widely adopted in regulatory contexts such as JARUS and SESAR [2, 3, 4, 8].

Despite their widespread use, these approaches exhibit important limitations. In particular, existing work predominantly focuses on risk evaluation, while comparatively little attention is given to safety verification during actual operation and to learning from operational feedback [2]. This gap highlights a fundamental limitation: while risks can be assessed a priori, ensuring safety under real-world conditions remains challenging, especially in the presence of uncertainty and disturbances.

Simulation-based evaluation is frequently employed to complement analytical methods. Tools such as Gazebo, USim [9], and Total Airspace and Airport Modeler (TAAM) [10] enable experimental testing of UAV behavior under controlled

conditions. However, the resulting safety assessments are inherently empirical and scenario-dependent, and therefore do not provide formal guarantees on system behavior. Moreover, commonly used safety metrics are largely focused on collision-related quantities. These include conflict probability, defined as the probability that the separation between two aircraft falls below a prescribed minimum threshold, and separation violations, which refer to events where this minimum safety distance is breached. Broader aspects of system reliability and robustness are less frequently addressed [2].

At the trajectory planning level, a wide range of methods has been proposed to generate feasible and efficient paths. Classical approaches include graph-based algorithms such as A* and D*, sampling-based planners such as RRT and RRT*, and optimization-based trajectory generation techniques. More recently, learning-based and data-driven methods have been introduced to improve performance in complex and dynamic environments [11]. While these approaches have demonstrated strong empirical performance, they typically lack formal safety guarantees and are often sensitive to parameter tuning, model inaccuracies, and distributional shifts. In particular, learning-based methods may degrade under unmodeled disturbances or sensor noise, potentially leading to unsafe behavior [11].

To address these limitations, risk-aware planning approaches have been proposed that incorporate probabilistic risk measures into the planning process. For example, Braßel et al. introduce a trajectory optimization framework that explicitly accounts for risk, while Chen et al. propose a multi-objective formulation that balances path efficiency and safety using a risk metric [12, 13]. However, these approaches treat safety primarily as an optimization objective rather than a formally verifiable constraint, and therefore do not provide explicit guarantees on collision probability or system reachability.

A more principled framework for safety analysis is provided by probabilistic reachability methods for nonlinear stochastic systems [7, 5]. These approaches characterize uncertainty propagation by constructing probabilistic bounds on system trajectories, enabling formal safety verification with high confidence. In particular, techniques based on probabilistic tubes or set erosion yield time-uniform guarantees on state deviations under stochastic disturbances [7, 5, 6]. However, despite their strong theoretical foundations, such methods have seen limited integration into practical UAV autonomy pipelines. As a result, current approaches either rely on structured but largely offline risk assessment or generate feasible trajectories without formal safety guarantees under uncertainty, leaving their integration into realistic planning and control frameworks an open challenge.

III. METHOD

This section presents the theoretical framework for probabilistic safety verification of UAV trajectories under stochastic disturbances, with a particular focus on wind-induced uncertainties generated by a simulation-based turbulence model. The theoretical formulation follows the framework of Liu et al., while explicitly grounding the stochastic disturbance model

in a physically motivated Dryden or von Kármán-based wind simulation used in the proposed WindPlugin [5].

A. Problem Formulation

A discrete-time stochastic dynamical system of the form

$$X_{t+1} = f(X_t, d_t, t) + w_t, \quad (1)$$

is considered, where $X_t \in \mathbb{R}^n$ denotes the system state, $d_t \in \mathcal{D} \subset \mathbb{R}^m$ is a bounded input or disturbance, and $w_t \in \mathbb{R}^n$ is a stochastic disturbance.

The corresponding deterministic system is given by

$$x_{t+1} = f(x_t, d_t, t), \quad (2)$$

which represents the noise-free evolution of the system. Pairs of associated trajectories (X_t, x_t) are defined as trajectories initialized at the same initial state and subject to the same input sequence.

The objective is to characterize the stochastic trajectory deviation

$$S_t := X_t - x_t \quad (3)$$

over a finite time horizon $t \leq T$ with high probability.

Standard assumptions on the system dynamics and disturbances are adopted. First, the function f is assumed to be Lipschitz continuous in the state. Therefore, there exists $L_t \geq 0$ such that

$$\|f(x, d, t) - f(y, d, t)\| \leq L_t \|x - y\|, \quad \forall x, y \in \mathbb{R}^n. \quad (4)$$

Second, the disturbance w_t is assumed to be sub-Gaussian with variance proxy σ_t^2 , meaning that it has zero mean and satisfies exponential tail bounds comparable to a Gaussian random variable.

Under these assumptions, the deviation between stochastic and deterministic trajectories can be bounded with high probability. In particular, there exists a function $r_{\delta, t}$ such that

$$\mathbb{P}(\|X_t - x_t\| \leq r_{\delta, t}, \quad \forall t \leq T) \geq 1 - \delta. \quad (5)$$

A key result from the set-erosion framework provides an explicit expression for this bound:

$$r_{\delta, t} = \sqrt{\Psi_t \left(\varepsilon_1 n + \varepsilon_2 \log \left(\frac{T}{\delta} \right) \right)}, \quad (6)$$

where Ψ_t depends on the Lipschitz constants and noise magnitude, and $\varepsilon_1, \varepsilon_2$ are constants determined by concentration inequalities. This bound grows logarithmically with the time horizon T and the inverse confidence level $1/\delta$, yielding a tight characterization of stochastic deviations [5].

B. Set-Erosion-Based Safety Method

Let $\mathcal{C} \subset \mathbb{R}^n$ denote the safe set. Direct safety verification of the stochastic system requires ensuring

$$\mathbb{P}(X_t \in \mathcal{C}, \quad \forall t \leq T) \geq 1 - \delta, \quad (7)$$

which is generally difficult to evaluate.

The set-erosion strategy provides a tractable sufficient condition. Specifically, define the eroded safe set

$$\tilde{\mathcal{C}}_t = \mathcal{C} \ominus \mathcal{B}(r_{\delta, t}), \quad (8)$$

where $\mathcal{B}(r_{\delta, t})$ is a ball of radius $r_{\delta, t}$ centered at the origin, and $\ominus$ denotes the Minkowski difference.

The set-erosion principle states that if the deterministic trajectory satisfies

$$x_t \in \tilde{\mathcal{C}}_t, \quad \forall t \leq T, \quad (9)$$

then the stochastic system is safe with probability at least $1 - \delta$.

This follows from the fact that, with high probability, the stochastic trajectory remains within a ball of radius $r_{\delta, t}$ around the deterministic trajectory. Ensuring that the deterministic trajectory stays within the eroded set therefore guarantees that the stochastic trajectory remains within the original safe set [5].

The above formulation enables direct integration into trajectory planning and control. Instead of explicitly propagating stochastic dynamics, trajectories can be planned for the deterministic system subject to tightened state constraints defined by the eroded set $\tilde{\mathcal{C}}_t$.

This reformulation shifts the stochastic safety verification problem into a deterministic one with modified constraints, allowing the use of standard tools.

The overall computational complexity is therefore dominated by deterministic planning, while the stochastic component is captured through analytically derived bounds that can be evaluated efficiently [5].

C. Disturbance Modeling

UAVs operating in real-world environments are subject to disturbances including sensor noise, communication delays, and atmospheric effects such as wind gusts and turbulence [14]. To reproduce realistic operating conditions, wind disturbances are explicitly modeled in the simulation.

Atmospheric turbulence is characterized as stochastic, small-scale fluctuations in the wind velocity field arising from pressure and temperature instabilities [15]. Following aerospace standards such as MIL-F-8785C and MIL-HDBK-1797, turbulence is represented as a stochastic process defined via its velocity power spectral density (PSD) [16, 17]. This formulation reflects the inherently statistical nature of atmospheric motion, which is not deterministically predictable.

A range of wind and turbulence models exists, varying in fidelity, computational cost, and application domain. Figure 1 provides an overview of commonly used approaches [18].

In this work, turbulence is modeled under standard assumptions: stationarity in time, spatial homogeneity, and isotropy. Additionally, turbulence length scales are assumed to be significantly larger than the UAV dimensions, implying spatially uniform excitation across the vehicle [15]. These assumptions are standard in flight dynamics simulation.

Among available models, the Dryden and von Kármán formulations are widely used in aerospace applications due to their PSD-based structure and computational efficiency [19,

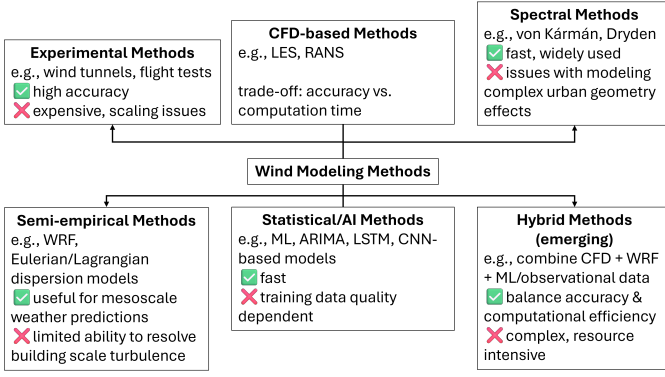


Fig. 1. Overview of wind and turbulence modeling approaches.

18, 20]. Both provide stochastic velocity fields suitable for real-time simulation.

While both models produce comparable aircraft response characteristics under turbulent excitation [15], their usage differs in practice: the von Kármán model is typically preferred for analysis due to its higher spectral fidelity, whereas the Dryden model is favored in simulation contexts due to its simpler rational transfer function structure [20].

Both formulations define turbulence in the frequency domain via PSD functions, which are used to generate correlated velocity fluctuations in the longitudinal, lateral, and vertical directions. The Dryden model is based on rational transfer functions, whereas the von Kármán model employs irrational spectral forms, providing improved physical accuracy at increased computational cost [20].

The one-dimensional Dryden spectra are given by:

$$\Phi_u(\Omega) = \sigma_u^2 \frac{2L_u}{\pi} \frac{1}{1 + (L_u\Omega)^2} \quad (10)$$

$$\Phi_v(\Omega) = \sigma_v^2 \frac{L_v}{\pi} \frac{1 + 3(L_v\Omega)^2}{[1 + (L_v\Omega)^2]^2} \quad (11)$$

$$\Phi_w(\Omega) = \sigma_w^2 \frac{L_w}{\pi} \frac{1 + 3(L_w\Omega)^2}{[1 + (L_w\Omega)^2]^2} \quad (12)$$

The corresponding von Kármán spectra are:

$$\Phi_u(\Omega) = \sigma_u^2 \frac{2L_u}{\pi} \frac{1}{[1 + (1.339L_u\Omega)^2]^{5/6}} \quad (13)$$

$$\Phi_v(\Omega) = \sigma_v^2 \frac{2L_v}{\pi} \frac{1 + \frac{8}{3}(1.339L_v\Omega)^2}{[1 + (1.339L_v\Omega)^2]^{11/6}} \quad (14)$$

$$\Phi_w(\Omega) = \sigma_w^2 \frac{2L_w}{\pi} \frac{1 + \frac{8}{3}(1.339L_w\Omega)^2}{[1 + (1.339L_w\Omega)^2]^{11/6}} \quad (15)$$

where Φ_u , Φ_v , and Φ_w denote the PSDs of the longitudinal, lateral, and vertical velocity components as functions of spatial frequency Ω [rad/m]. The parameters σ_u , σ_v , and σ_w define root-mean square (RMS) turbulence intensities, while L_u , L_v ,

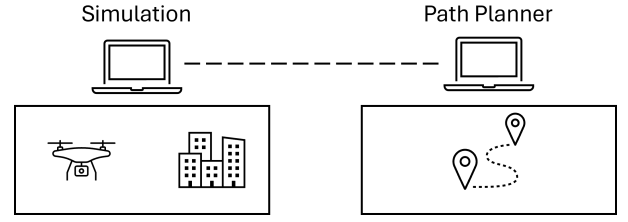


Fig. 2. Simulation architecture showing a distributed setup with two computing units: one running the Gazebo–ArduPilot UAV simulation environment, and the other executing the external path planning module that transmits waypoint trajectories to the simulation computer via MAVLink.

and L_w represent the corresponding integral length scales governing spatial correlation [20].

The spatial frequency Ω is related to temporal frequency via the mean flow velocity under the frozen turbulence assumption. The constant 1.339 in the von Kármán formulation is empirically derived to match observed atmospheric energy spectra [20].

For low-altitude conditions (below 1000 ft above ground level (AGL)), military standards such as MIL-F-8785C and MIL-HDBK-1797 define turbulence parameters as altitude-dependent functions. In particular, vertical length scales are often assumed to scale approximately linearly with altitude, while longitudinal and lateral scales follow empirical functions of height h . Turbulence intensities are commonly parameterized using reference wind speed at 20 ft altitude (W_{20}), with directional scaling relationships linking σ_u , σ_v , and σ_w [20].

The resulting velocity field is expressed in a wind-aligned coordinate system and subsequently transformed into the UAV body-fixed frame for use in flight dynamics simulation. This enables generation of temporally and spatially correlated stochastic wind fields used as disturbances in the simulation environment.

IV. EXPERIMENT SETUP

The theoretical developments from Section III are implemented within a physics-based simulation environment. The following subsections outline the system model, disturbance representation, and planning methodology used in the experiments.

A. System Model and Coordinate Frames

The simulation environment is implemented in Gazebo Harmonic running on Ubuntu 24.04 (Noble) and is integrated with ArduPilot via the Gazebo–ArduPilot bridge. All experiments are conducted using the gazebo-iris multirotor platform as the underlying UAV model.

The overall system is distributed across two computational units. The first unit executes the physics-based UAV simulation, including vehicle dynamics, sensor simulation, and autopilot logic via Gazebo and ArduPilot. The second unit runs the external path planning module, which computes collision-free trajectories based on a predefined obstacle configuration.

Communication between both components is realized using the MAVLink protocol. The planner transmits sequences of

spatial waypoints to the autopilot, which tracks the reference trajectory using its onboard control system.

The implemented path planning framework integrates deterministic grid-based motion planning with stochastic safety margins derived from probabilistic reachability analysis in the theoretical section. The objective is to generate collision-free trajectories that remain feasible with a probability of at least $1 - \delta$ under bounded stochastic disturbances and model uncertainty.

The planning process is formulated in a discretized two-dimensional spatial domain defined in the East-North-Up (ENU) coordinate frame. This representation is chosen for consistency with the geometric structure of the environment and for straightforward integration of obstacle geometry.

All control commands executed by the autopilot are expressed in the North-East-Down (NED) frame, in accordance with the MAVLink interface specification. Consequently, a consistent coordinate transformation between ENU and NED frames is maintained throughout the planning and execution pipeline.

B. Stochastic Disturbance Model

Atmospheric disturbances are incorporated into the simulation using a custom Gazebo plugin, *AdvancedWindSystem*, which generates stochastic wind fields acting on the UAV. The plugin implements either a Dryden or von Kármán turbulence model, as described in Section III-C, and produces a three-dimensional wind velocity perturbation $\boldsymbol{\eta} = (u, v, w)$ expressed in the UAV body-fixed frame.

Each velocity component is modeled as a zero-mean, wide-sense stationary Gaussian process, implying that the turbulence exhibits no deterministic bias and that its statistical properties, such as variance and correlation structure, remain constant over time. The intensity of the fluctuations is characterized by the RMS values $\sigma_u, \sigma_v, \sigma_w$, while the spatial correlation lengths L_u, L_v, L_w define the rate at which the wind field varies spatially.

The turbulence models are realized in discrete time as linear shaping filters driven by unit-variance white Gaussian noise $\xi(t) \sim \mathcal{N}(0, 1)$, such that the resulting output reproduces the prescribed velocity PSD. The RMS turbulence intensity is parameterized as $\sigma = I \cdot V_{\text{ref}}$, where I denotes the turbulence intensity factor and V_{ref} is the reference wind speed. The resulting signal $\boldsymbol{\eta}(t)$ represents the instantaneous turbulent wind velocity relative to the UAV.

Aerodynamic disturbances are modeled as a quadratic drag force acting in the body frame:

$$\mathbf{F}_{\text{body}} = \frac{1}{2} \rho C_d A |\boldsymbol{\eta}| \boldsymbol{\eta}, \quad (16)$$

where $\rho = 1.225 \text{ kg/m}^3$ is the air density and $C_d A = 0.2 \text{ m}^2$ denotes the effective drag area. The computed force is applied to the UAV dynamics via the Gazebo *ExternalWorldWrenchCmd* interface, with the configuration parameter `apply_to_all_links` controlling whether

the force is distributed across all links or applied to a single reference link.

To avoid transient initialization effects, wind excitation is activated only above a minimum altitude threshold of `min_altitude = 2.0m`. For monitoring and analysis, the instantaneous wind vector is additionally published on the ROS 2 topic `/sim/wind`.

C. Path Planning under Stochastic Disturbances

The environment is discretized into a uniform two-dimensional occupancy grid with resolution-dependent cell size, resulting in a finite graph representation of the configuration space used for path search.

Prior to grid construction, obstacles are inflated to account for both the UAV geometry and stochastic deviations induced by disturbances. This inflation is formulated as a Minkowski sum between the obstacle set and a safety set derived from the UAV footprint and the stochastic deviation bound obtained in Section III. The resulting inflation radius combines the physical UAV radius with the worst-case uncertainty growth over a finite horizon under Lipschitz dynamics and sub-Gaussian disturbances.

This procedure yields a reduced free space in which any feasible trajectory satisfies the probabilistic safety constraint with confidence level $1 - \delta$.

A graph-based A* search is then applied to compute a collision-free path from the initial state to the goal. The cost function is defined by Euclidean distance in grid space, and an admissible Euclidean heuristic ensures optimality with respect to the discretized graph.

The resulting path is simplified by removing collinear nodes and further refined through a line-of-sight shortcutting procedure, replacing multi-segment subpaths with direct connections whenever collision-free visibility is preserved in the inflated occupancy grid.

The discrete trajectory is subsequently mapped to continuous space by transforming grid coordinates into world coordinates and assigning a constant flight altitude in the NED frame. This yields a sequence of waypoints representing the nominal UAV trajectory.

By incorporating the stochastic deviation bounds directly into obstacle inflation, uncertainty is handled at the planning stage through a modification of the free space. This avoids the need for explicit trajectory tube representations while ensuring probabilistic safety guarantees.

During execution, the UAV tracks the waypoint sequence using MAVLink position control commands. Each waypoint is maintained until the position controller converges within a specified tolerance before advancing to the next, resulting in a piecewise-constant approximation of the continuous reference trajectory consistent with the discrete-time control interface.

V. RESULTS

All experiments were conducted on a system equipped with an AMD Ryzen processor and 14 GB of RAM. As established

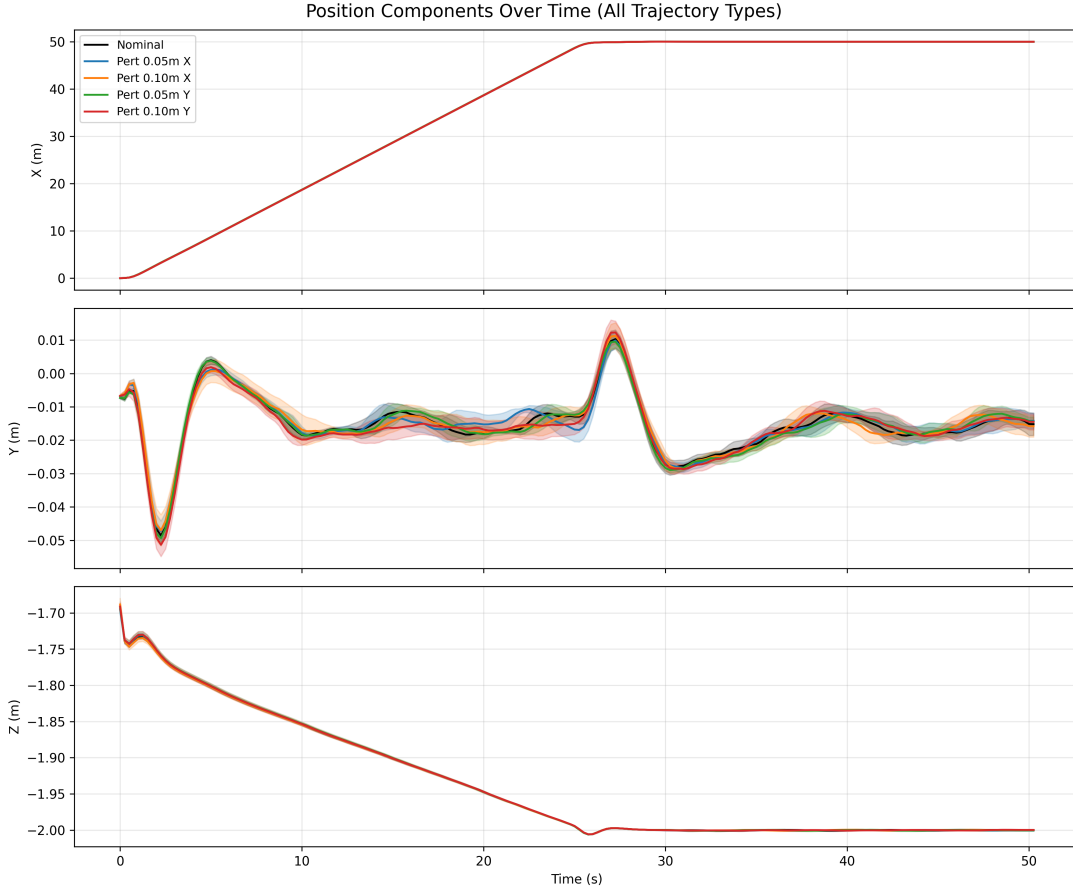


Fig. 3. Mean trajectories and standard deviations for all perturbation cases compared to the nominal trajectory.

in Section III, the validity of the proposed probabilistic trajectory bounds relies on contractive closed-loop system behavior. Accordingly, the UAV dynamics are evaluated empirically under perturbed initial conditions.

A. Trajectory Convergence Analysis

Five trajectory configurations are considered, as summarized in Table I. The nominal initial position is defined as $(0, 0, 0.195)$ and configured in the simulation world file `iris_runway.sdf` (Listing 1). Perturbed trajectories are generated by introducing small offsets in either the x - or y -direction.

Listing 1. Unperturbed UAV Position

```

1 <include>
2   <uri>model://iris_with_gimbal</uri>
3   <pose degrees="true">0 0 0.195 0 0 90</pose>
4 </include>

```

Figure 3 illustrates the mean trajectories and corresponding standard deviations for all perturbation cases. Despite the initial offsets, all trajectories converge toward a common nominal path, indicating stable closed-loop behavior.

A slightly increased dispersion is observed in the y -direction. Given the small perturbation magnitudes (± 0.1 m),

TABLE I
TRAJECTORY TYPES

Trajectory Type	Initial Position
Nominal	$(0, 0, 0.195)$
005_x	$(0.05, 0, 0.195)$
01_x	$(0.1, 0, 0.195)$
005_y	$(0, 0.05, 0.195)$
01_y	$(0, 0.1, 0.195)$

this effect is primarily attributed to scaling in the visualization rather than a significant difference in system response.

To quantify convergence, the Euclidean deviation between perturbed and nominal trajectories is evaluated over time. The results are shown in Figure 4.

The tracking error decreases monotonically after an initial transient phase and converges toward zero for all configurations. This behavior confirms bounded tracking error and supports the assumption of contractive system dynamics.

Perturbations in the y -direction consistently yield larger errors, suggesting directional asymmetry in the closed-loop response. This may be attributed to differences in controller tuning or vehicle dynamics along the lateral axis.

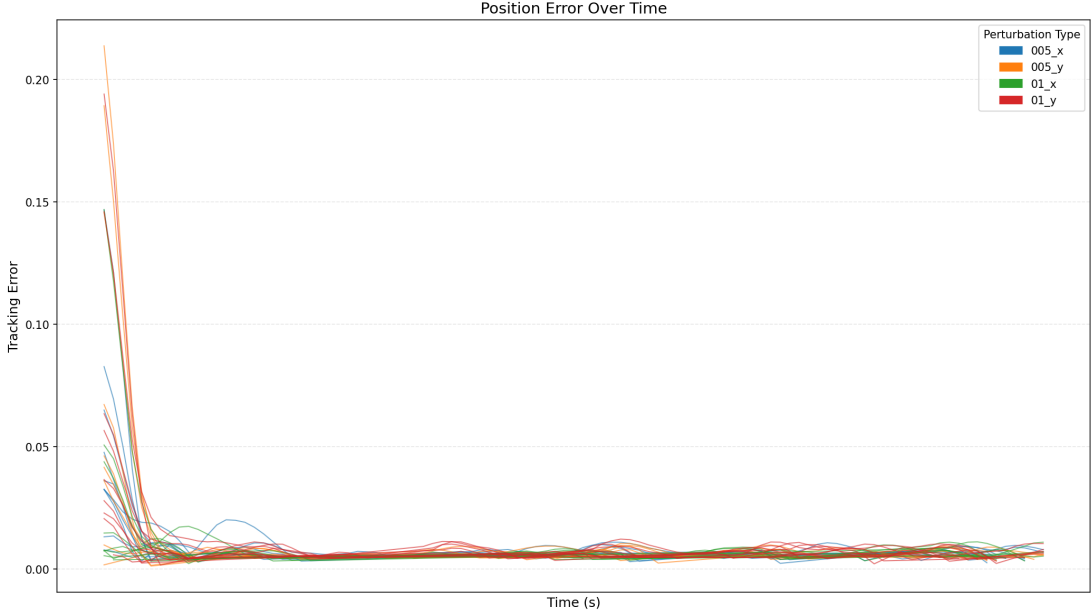


Fig. 4. Trajectory tracking error as a function of time for all perturbation cases. The initial transient phase is omitted to highlight steady-state convergence behavior.

TABLE II
MEAN TRACKING ERROR STATISTICS

Trajectory Type	Mean Error [m]	Standard Deviation [m]
005_x	0.0282	0.0206
01_x	0.0172	0.0217
005_y	0.0424	0.0403
01_y	0.0327	0.0313

TABLE III
STATISTICAL SUMMARY OF LIPSCHITZ CONSTANT ESTIMATES

Type	N	Mean	Median	GeoMean	% $L \leq 1$
0.05_x	1121	1.009	0.977	0.869	57.18
0.05_y	1071	0.980	0.983	0.856	58.64
0.1_x	960	1.126	0.949	0.858	56.88
0.1_y	1216	1.005	0.971	0.877	57.15

B. Contractivity Analysis

The Lipschitz constant is estimated empirically from time-aligned nominal and perturbed trajectories using a finite-difference approximation. Let $e_k = \|x_k - \tilde{x}_k\|$ denote the Euclidean deviation between corresponding states at time t_k . A time-varying Lipschitz factor is then computed as

$$L_k = \exp\left(\frac{\log e_k - \log e_{k-1}}{t_k - t_{k-1}}\right),$$

which corresponds to the exponential growth rate of the trajectory deviation.

To ensure numerical robustness, samples within an initial transient phase are excluded, and ratios involving small deviations are discarded. The reported statistics are computed over all valid samples across multiple runs and perturbation configurations.

The statistical summary is provided in Table III.

Across all configurations, the estimated Lipschitz constants are concentrated around unity, indicating operation near the boundary of contractivity. Median and geometric mean values remain consistently below one, suggesting that the system exhibits predominantly contractive behavior.

The distributions are right-skewed, with occasional large values increasing the mean. These outliers are likely caused

by numerical effects, such as division by small denominators, rather than reflecting intrinsic instability. This interpretation is supported by the convergence of trajectory errors and the high proportion of samples satisfying $L \leq 1$.

C. Trajectory Execution in Simulation

The experimental scenario is illustrated in Figure 5. Obstacles are modeled as vertical structures with a height of 15 m, while the UAV operates at a constant altitude of 5 m. This setup effectively reduces the planning problem to two dimensions without compromising collision avoidance.

The UAV is initialized at (0,0) and tasked with reaching a landing pad at (25,25). After simulation startup, the external planner computes a collision-free path based on the inflated obstacle representation.

Figure 6 shows the obstacle inflation resulting from the stochastic safety margin. The total inflation radius is 3.343 m, combining the UAV physical radius (0.33234 m) and the disturbance-induced safety margin (3.010 m).

The resulting trajectory, shown in Figure 7, consists of five waypoints in the NED frame:

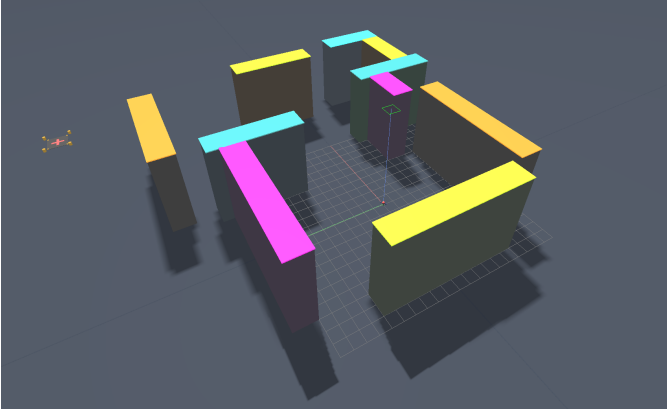


Fig. 5. Simulation environment with obstacle configuration and UAV mission setup.

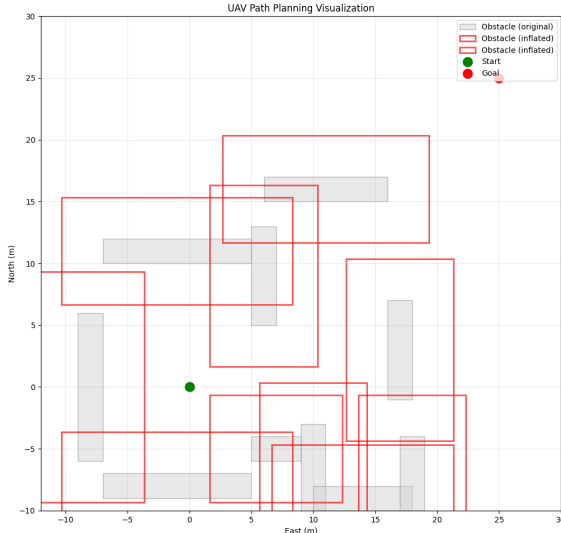


Fig. 6. Obstacle inflation based on stochastic safety margins.

(0.25, 0.25, -5.00),
 (1.25, 10.75, -5.00),
 (10.75, 12.25, -5.00),
 (11.25, 19.75, -5.00),
 (25.00, 25.00, -5.00)

The UAV successfully tracks the waypoint sequence and reaches the goal location. The executed trajectory closely follows the planned path, demonstrating consistency between planning and control.

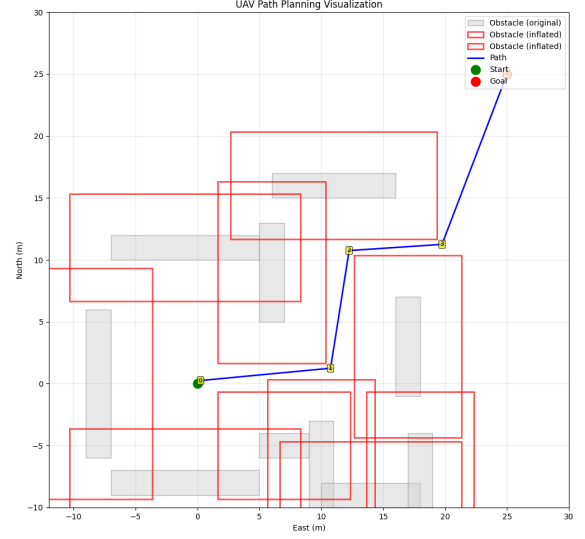


Fig. 7. Planned UAV trajectory in the inflated environment.

D. Effect of Wind Disturbances

During execution, deviations from the nominal trajectory are observed due to stochastic wind disturbances. These disturbances are generated by the custom wind plugin described in Section IV-B.

Wind excitation is enabled only above an altitude of 2 m to avoid interference with takeoff dynamics. The wind model is configured with a reference speed of 5.0 m/s and a turbulence intensity of 0.5, resulting in significant stochastic fluctuations around the mean flow.

The Dryden model is used with a scale length of 10.0, producing spatially correlated wind fields. Disturbance forces are applied to the main body link to ensure stable simulation behavior.

Figures 8 and 9 show the resulting UAV trajectories under wind disturbances.

While the disturbances introduce visible deviations, the UAV remains within the safe corridor defined by obstacle inflation and successfully completes the mission. This demonstrates the effectiveness of the proposed planning approach in maintaining probabilistic safety under realistic environmental disturbances.

VI. FUTURE WORK

While the proposed framework demonstrates effective probabilistic safety verification for UAV trajectory planning under stochastic disturbances, several directions remain open for further research and improvement. A natural extension of the current work is the transition from a predominantly two-dimensional planning formulation to a fully three-dimensional environment. In the present implementation, the UAV operates

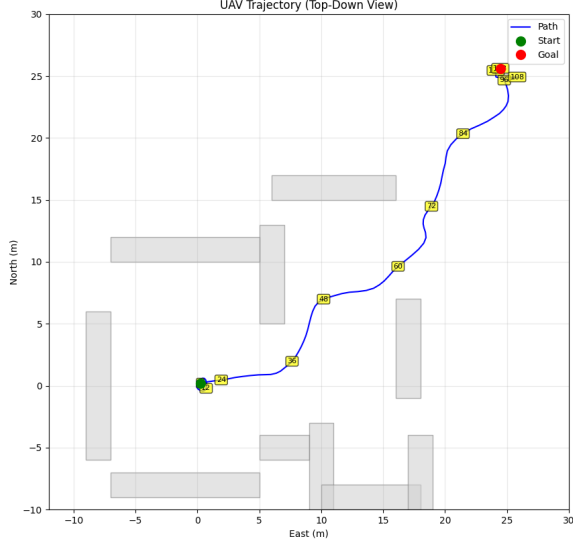


Fig. 8. UAV trajectory in the horizontal plane under wind disturbances.

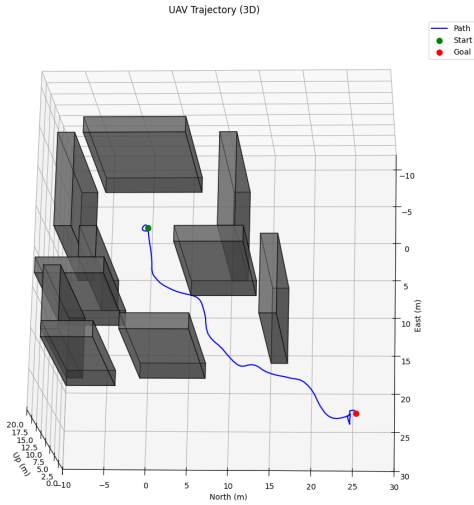


Fig. 9. Three-dimensional UAV trajectory under wind disturbances.

at a fixed altitude, which simplifies the collision avoidance problem to a planar setting. Extending the framework to full 3D motion would allow altitude to be explicitly incorporated as an additional planning dimension, enabling more realistic mission scenarios such as urban airspace navigation, indoor flight, and operations in cluttered vertical environments. This extension would require generalizing the set-erosion formulation to volumetric obstacle representations and adapting the Minkowski-based safety margins to three-dimensional geometry.

Another important direction concerns the modeling of

environmental disturbances. The current approach relies on standard Dryden and von Kármán turbulence models, which assume spatial homogeneity and isotropy of the wind field. While these assumptions are well-established in aerospace simulation, they do not fully capture the complexity of real-world environments, particularly in urban areas where wind behavior is strongly influenced by buildings, terrain, and localized flow effects. Future work should therefore focus on integrating more realistic and spatially varying disturbance models, potentially derived from high-fidelity computational fluid dynamics simulations or data-driven approaches trained on empirical wind measurements. Such models would allow a more accurate representation of the uncertainty affecting UAV motion in practical deployments.

In addition, the probabilistic safety bounds derived in this work are based on fixed assumptions regarding Lipschitz continuity and disturbance statistics. While this leads to analytically tractable expressions, it introduces conservatism and does not account for temporal variations in system behavior or environmental conditions. A promising extension would be the development of adaptive or time-varying safety bounds that can be updated online based on observed system performance or real-time estimates of disturbance intensity. This would allow the framework to dynamically adjust the level of conservatism, potentially improving performance while still maintaining formal safety guarantees.

The current formulation is also limited to single-agent scenarios operating in static obstacle fields. Extending the framework to multi-UAV systems would introduce additional challenges arising from coupled dynamics and mutual interactions between agents. In such settings, each UAV contributes to the stochastic environment of the others, requiring distributed or decentralized formulations of probabilistic safety guarantees. Developing scalable multi-agent extensions would significantly increase the applicability of the approach to coordinated missions such as swarm-based inspection, search and rescue, or cooperative transport tasks.

Finally, although the proposed method has been validated in a high-fidelity simulation environment, experimental validation on real UAV platforms remains an important next step. Real-world deployment would allow assessment under true aerodynamic effects, sensor noise, and actuator imperfections that are difficult to fully capture in simulation. Such experiments would also provide valuable insight into the conservatism of the derived probabilistic bounds and their robustness under modeling mismatch.

Overall, these directions aim to further bridge the gap between theoretical probabilistic safety guarantees and their practical deployment in complex, uncertain, and dynamic real-world UAV applications.

VII. CONCLUSION

This work addressed the problem of ensuring safe UAV operation under stochastic disturbances by introducing a probabilistic safety verification framework that bridges the gap between formal analysis and practical trajectory planning.

While conventional approaches either rely on offline risk assessment or simulation-based validation, they typically do not provide formal guarantees on safety during execution. The proposed method overcomes this limitation by integrating probabilistic reachability analysis with deterministic planning through a set-erosion-based formulation.

By modeling UAV dynamics as a stochastic system with sub-Gaussian disturbances, high-probability bounds on trajectory deviations were derived and incorporated into the planning process via obstacle inflation. This allowed safety constraints to be enforced implicitly by reducing the feasible configuration space, enabling the use of standard graph-based planning algorithms without modification. The resulting framework is computationally efficient and modular, with the stochastic component admitting closed-form bounds and the overall complexity dominated by deterministic reachability and path planning.

The approach was validated in a Gazebo–ArduPilot simulation environment with realistic wind disturbances modeled using Dryden and von Kármán turbulence formulations. Experimental results demonstrated bounded tracking error, convergence toward nominal trajectories, and contractive system behavior, supporting the validity of the underlying assumptions. The successful execution of planned trajectories in the presence of stochastic disturbances highlights the practical applicability of the method.

Overall, this work shows that probabilistic safety guarantees can be integrated into UAV trajectory planning in a tractable and implementation-friendly manner. It provides a step toward bridging the gap between theoretically rigorous safety verification methods and real-world autonomous UAV systems operating under uncertainty.

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